

# Caffevolve — Phone-Primary Personalization

Addendum to “The Intelligence Center as a Distributed Process”: how resident profiles are reserved, finished, paired and kept in sync

**Prepared for:** Brent Langevin **Purpose:** Define which personalization tasks happen on the phone and which at the device, how a resident profile is reserved and paired, how edits stay in sync, and the commitments that keep the device useful on its own. **Status:** working draft, revised after review. The rules below were agreed in review; section 15 lists what is still open.

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## 1. The core rule, in one line

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**The phone is the authoring surface; the device is the execution surface.** Heavy personalization is done where the user is comfortable, and both surfaces may adjust what exists. All brew, grind and drink-path decisions, and every brew initiation, remain at the device.

## 2. Why

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Personalization has grown: the initial profile, the Caffeine Behavior Instrument (CBI), subprofiles, BrewNow profiles, additive and grind preferences, and now daily plans and routines. Doing all of it at the device means long sessions standing at a countertop. Moving the heavy, optional authoring to the phone removes that friction and should reduce reluctance to use the personalization tools. The architecture already supports it: the engine logic is compiled into the app, and the per-user cloud store already carries the event log and the user’s parameters between the two copies of the engine.

## 3. Two stages of a resident profile

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A Resident User Profile is built in two stages, by two different people.

| Stage                                                         | Who                                     | Where                                   | What it holds                                                                                                                                                 |
|---------------------------------------------------------------|-----------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Basic Resident Profile</b>                                 | The administrator, behind the Admin PIN | At the device (Administrator Functions) | First name, initials, profile color and the user’s <b>email address</b> , which becomes the account’s user ID. It reserves one of the device’s profile slots. |
| <b>Profile setup, producing the Complete Resident Profile</b> | The user, with no administrator PIN     | Device or phone                         | The CBI and the ceilings it derives, then the optional layer described in section 5.                                                                          |

- **Every resident profile starts this way, including the administrator's own.** The administrator first sets up the device (connection, beans, additives and the Admin PIN). No profile exists yet. The administrator then creates their own Basic Resident Profile, at that time or later, by the same route as everyone else.
- **The administrator never answers the CBI for anyone.** The slot is reserved by the administrator; everything personal is completed by the user.
- **Until the user finishes,** the profile appears as “Ready for you” with a “Finish setup” action, and the resident brewing paths (Home, Away, Plan my Day) stay locked. The user can brew through any guest channel in the meantime. The user may pause and resume setup, but cannot discard the reserved slot. Only the administrator removes a resident profile.
- **Resumable.** Setup already offers “Exit setup & complete later.” A user may answer part of the CBI on one surface and finish on either. The profile is established only when the CBI completes.
- **Bedtime is not part of the profile.** It is a session value, set and revisited at the device with each brew. It is never synced and cannot conflict.
- **Slots** are limited by tier (Originara 2, Advancara 5, Culminara Office 20), so the administrator decides who holds one.

## 4. Reserving, inviting and the account

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- **The email address is part of the administrator's step.** It is entered with the name, initials and color. The invite code is sent to it, and it becomes the user's account ID when the code is redeemed. It is not a credential. No phone number is collected anywhere.
- **The code.** On completing the Basic Resident Profile, the device registers the shell (name and email address) with the cloud and requests a one-time code, which is emailed to that address with a link to download the app. The device must be online at that moment. The user's personal data (CBI answers, ceilings, preferences) stays on the device until they pair.
- **The code is one-time, expires after a short window, locks after a few wrong attempts, and binds to the first account that redeems it.** The administrator can issue a fresh code, correcting the address if needed, if one expires or is lost.
- **Cleanup.** If a code expires unused, the shell's email address is deleted from the cloud. When the code is redeemed, the address becomes the account ID.
- **The account.** Redeeming the code and downloading the app creates the account. The invited email address is the user ID, and it is verified by the redemption itself. The user sets a password. From then on, the email and password alone give access. The user then finishes the profile on the phone.
- **Recovery.** A forgotten password is reset by email, with no administrator involved. The password has minimum rules, and repeated failed sign-ins lock the account temporarily. A new or second phone is handled by signing in with the same email and password. Optional

phone biometric unlock is a convenience after the first sign-in.

- **One account, one profile, bound permanently.** The claim cannot be undone or transferred, and there is no reassign function. The same profile works at home and at the office. If the wrong person claims a profile, for example because the address was mistyped, the administrator removes the profile and creates a new Basic Resident Profile. Removal deletes the cloud copy and unpairs the account.
- **Joining a second device.** The second device's administrator reserves a slot with the person's name, initials, color and the email address of their existing account, and issues the usual invite. The person redeems it while signed into that account, which attaches their existing profile to that device. It does not create a second profile, and no new CBI is needed. Each device an account uses consumes one slot on that device. The invited address must match the account's email.
- **An unpaired profile works fully at the device.** A local-only profile has no cloud backup, and a device reset or failure loses it. The app and the device say so when offering the invite.
- **Phone-first.** A person cannot self-create a resident profile on a household device. The administrator-issued invite is the only route to a slot, so one mechanism covers both directions.

## 5. What is created where

| Item                                                                                                  | Create                                                    | Edit                                                                                                                   |
|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Basic Resident Profile: name, initials, color, email address (becomes the account ID)                 | Administrator, at the device (Admin PIN)                  | As today                                                                                                               |
| Complete Resident Profile: CBI and the ceilings it derives                                            | User, device or phone. Phone preferred. Resumable.        | Either, through the profile review screen                                                                              |
| Optional layer: additive preferences, grind setting, subprofiles, BrewNow profile, workplace settings | Phone only                                                | Either                                                                                                                 |
| Today's plan (Plan my Day): build, or resume a routine for today                                      | Either surface. Lasts for that day only and is not saved. | Either surface: add, change or remove any drink (time, type, amount), protect a drink, run Rebalance, or skip the plan |
| Routines (saved plans): create, edit, name, set saved days, delete                                    | Phone only                                                | Phone only                                                                                                             |
| Guest profile (Quick Profile Path)                                                                    | Device only                                               | Device only                                                                                                            |

- **Complete on the device.** A resident can finish the CBI and ceilings at the device with no phone and no account. The device is never unusable for want of a phone. The device first-run screen says the longer setup is faster on the phone and offers to continue there.

- **Phone-only layer.** On the device, screens for the optional layer say “available in the CaffeolveGo app” and do not appear broken. Until the user sets them, the defaults are **no additives and a medium grind**.
- **Today’s plan versus routines.** A routine is a saved plan, kept for the days the user chooses (its **saved days**). Adjusting today’s plan never writes back to the routine. The device never saves or changes a routine, and runs Plan Check against whichever plan is in effect today.
- Each item is created in one place at a time, which removes most conflict risk. Remaining conflicts occur only on edits, covered in sections 7 and 8.

## 6. The boundary model

The Caffeine Comfort Zone has four parameters. Users can adjust only the two ceilings.

| Parameter           | Source                       | User-adjustable                                                      |
|---------------------|------------------------------|----------------------------------------------------------------------|
| Wake-Hours Ceiling  | CBI, then refined by the AFS | Yes, any time, from the profile review screen (“Fine-tune ceilings”) |
| Bedtime Ceiling     | CBI, then refined by the AFS | Yes, same as above                                                   |
| Peak-Zone Floor     | CBI, then refined by the AFS | No                                                                   |
| Caffeine Decay Rate | CBI, then refined by the AFS | No                                                                   |

- **Each ceiling has three parts.** The *computed value* (from the CBI, moved by the AFS); an optional *user override* (set with the slider); and the *effective value*, which is the override if one exists, otherwise the computed value. The device brews against the effective value. “Use computed” removes the override.
- **The platform only advises.** Setting a ceiling higher than most people are comfortable with shows an advisory message, not a block. The user decides.
- **The AFS never overwrites an override.** It continues to refine the computed value in the background, so “Use computed” returns the learned value. Its boundary-triggered prompts fire around the effective value.
- **Refinement is stored as feedback events.** Each feedback response is an immutable event with a source and ID, and the computed value is derived from the baseline plus those events. Events arrive from either surface and merge as a simple union, in any order, so the two surfaces cannot disagree about the computed value once they hold the same events and the same engine version.

## 7. Editing and conflicts

The existing principle carries over: state changes are appended, not edited in place. Applied to profiles, subprofiles, preferences and plans:

- Every edit produces a **new version of the record**, stamped with its source (phone or device), time and the version it was based on.
- If both surfaces edited from the same base version, the **later edit wins per field**. A ceiling change on one side and a rename on the other both survive.
- If both changed the **same field**, the later edit wins and the other surface shows a one-line notice that its earlier change was replaced.
- The **two ceilings are treated as a group**, because they are adjusted together. If both surfaces edited them, the later edit wins for the pair, so the result is never a mix the user did not see.
- Each surface shows **sync state** (for example, “Synced to your Caffevolve 2 min ago”), especially after a ceiling change, because the device brews against whatever it last received.

For brew events the device remains the source of truth. For personalization records, the most recent valid version is authoritative, whichever surface wrote it.

## 8. Edit Profile: re-running setup

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From the profile review screen, **Edit Profile** re-runs the whole setup flow, and everything in it can be changed: the CBI answers, subprofiles, grind setting and additive preferences. It is available on either surface. A re-run has two kinds of outcome:

- **CBI answers unchanged.** Only the other fields change, and they merge as ordinary field edits. The user’s ceiling overrides and the AFS learning are kept. Changing an additive must not wipe tuned ceilings.
- **CBI answers changed.** This is a new baseline. The computed boundaries and decay rate are recomputed, the user’s **ceiling overrides are cleared**, and the **AFS learning starts over**. The user passes through the fine-tune screen again and may set new overrides.
- Any edit made on the other surface against the old baseline is superseded by the new baseline, and that surface shows a notice.

## 9. Plan my Day and bedtime

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Two different bedtimes are involved, and they are not the same thing.

- **Planned bedtime** is a required input when a plan or routine is set up. It lives with the plan, not the profile, and it is what the planner projects against.
- **Session bedtime** is set and confirmed at the device when a brew begins. It is not part of the profile. The session bedtime set at the device always rules over the planned bedtime.
- **The plan is re-projected first.** When a session bedtime is set, the device re-projects today’s plan against it before anything else, whether the new bedtime is earlier or later. Drinks already consumed stay locked as the baseline; only the drinks not yet consumed are re-evaluated. The projection runs again every time a session bedtime is set.

- **Earlier bedtime:** the bedtime residual rises, and if it now exceeds the Bedtime Ceiling the planner offers Rebalance on the remaining drinks. **Later bedtime:** the residual falls, and the planner shows the new bedtime residual and the headroom.
- **The plan is never changed automatically.** The platform advises and the user decides. The session bedtime applies to today's plan only and never writes back to a saved routine's planned bedtime. The in-path reminder (FIG. 43D) includes the bedtime change in its planned-versus-current delta.

## 10. Guests

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The phone-first rules apply to Resident Users only. The Guest path stays entirely on the device: name, initials, self-selected sensitivity level and bedtime, with no account, no invite and no CBI. A Returning Guest Profile continues to be kept for 72 hours.

## 11. Safety and access

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- **The Admin PIN stays, and is unchanged in purpose.** It secures the device and decides who holds a profile on it. It covers device configuration (network, beans, canisters, hardware), reserving a slot, issuing an invite code, removing a resident profile, and resetting or transferring the device (which wipes local profiles and unpairs the device). One entry per visit to Administrator Functions.
- **The profile PIN** exists only for the shared Culminara Office device: a user-chosen four-digit PIN required when a profile is selected on an Office device. It only gates access to an Office device, and is never required on a home device, even though the same profile is used in both places. Profile changes made from the phone are made in the user's account with email and password, so it never applies there. The two PINs protect different things and never overlap.
- **Profile edits need no extra PIN.** Adjusting a ceiling, changing additives or re-running setup follows the device's rule: nothing at home, the profile PIN on an Office device.
- The platform continues only to advise. The user decides what to brew and what their ceilings are.

## 12. Design commitments

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- **Minimal on-device path.** The device always supports the Complete Resident Profile (CBI and ceilings) and the Guest path, so it works without a phone.
- **Full local copy.** The device holds a complete copy of the active profiles and plans, so a cloud or app outage never strands a user's setup. Brewing continues from the last synced state.
- **The cloud remains a store, not a brain.** It holds the record and the current versions. It does not compute residual or decide what to brew.
- **Data governance defined before launch.** CBI answers describe sleep and caffeine habits and are health-adjacent, and the email address is personal data entered by one person about

another. It is held in the cloud as the account ID once the invite is redeemed, and deleted if the invite expires unused. Consent, retention, export and account deletion rules need to be set, in addition to per-user isolation and encryption.

- **Version matching extended.** The requirement that firmware and app carry the same engine version also covers the record schema and the shared personalization flows, so older firmware does not meet newer app flows unprepared.
- **Validate the assumption.** Test completion rates for the CBI and planning on a phone prototype against the device before committing the full build.

## 13. Prototype and reference changes this requires

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- Add an **email address field** (not a phone number) to the administrator’s “Set Up a New User” step and to the first-profile step, noting that it becomes the account ID.
- Add an **invite-code step** after the Basic Resident Profile is created, with an on-screen confirmation that a code was sent.
- Add the app-side screens: redeem the code, create the account (email as user ID, password, email verification), password reset, and the profile setup and review screens.
- Add a **join another device** path: an existing account redeems an invite and attaches its profile to the device.
- Change the device setup flow so the optional layer (additive, grind, subprofiles, BrewNow) is created in the app, with an “available in the CaffeolveGo app” message at the device.
- Update FIG. 43A step 3802: the profile load covers only the ceilings and decay rate, the **planned bedtime** is entered in the planning workspace, and a step re-projects the plan when a session bedtime is set at the device.
- Update FIG. 43A and the reference: “assign weekdays” (3832) and “matching the current weekday” (4106) become **saved days**, and the device’s Plan my Day allows adding, changing and removing drinks in today’s plan but offers “save as a routine” only in the app.
- Reconcile the reference text: the first profile is created by the administrator like any other, one account holds one profile used at home and at work, and the profile PIN applies to Office devices only.

## 14. Decisions recorded in review

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- The session bedtime set at the device always rules over a plan’s planned bedtime, and the plan is re-projected whenever it is set. Today’s instance only.
- The Peak-Zone Floor and Caffeine Decay Rate are not user-adjustable. Defaults before the user sets them on the phone: no additives, medium grind.
- Today’s plan can be built and adjusted on either surface and is not saved. Routines are managed on the phone only.

- A profile is never moved to a different account. One account holds one profile, used at home and at the office.
- The user ID is the account email, entered by the administrator and verified when the user redeems the invite. A password reset goes by email, with no administrator. The platform holds no phone number.
- No profile PIN on home devices; the Admin PIN stays.
- The ceiling overrides and the AFS reset only when the CBI answers change.
- The AFS never overwrites a user's override.

## 15. Open items

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- **Deferred to later versions:** password and lockout rules and any second sign-in factor; device resale and transfer (reset, detaching a device from accounts, a “My devices” list).
- Whether a later session bedtime should lead the planner to offer restoring strength to drinks that were eased by an earlier Rebalance (not decided).
- Whether a bedtime declared in CaffevolveGo counts the same way as one declared at the device (the most recent declaration winning).
- Email invite delivery: bounced or filtered messages, and the length of the invite expiry window.
- Whether profile color and initials may differ between a person's home and office devices.
- The prototype does not yet model AFS learning state, sync or pairing; these are specified here but not yet shown.

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*This addendum records the direction as agreed in review. Items in section 15 are not yet settled.*